

SEÇÃO Wick Wick Wick

$$\begin{aligned} \mathbb{E}[Z_{i_1 \alpha_1} Z_{i_2 \alpha_2}] &= \langle\langle Z_{i_1 \alpha_1} Z_{i_2 \alpha_2} \rangle\rangle_g + \frac{1}{8} \sum_{\beta_1, \beta_2, \beta_3, \beta_4 \in D} \sqrt{g^{\beta_1 \beta_2} g^{\beta_3 \beta_4}} \sum_{j_1, j_2} [\langle\langle Z_{j_1 \beta_1} Z_{j_1 \beta_2} Z_{j_2 \beta_3} Z_{j_2 \beta_4} Z_{i_1 \alpha_1} Z_{i_2 \alpha_2} \rangle\rangle_g \\ &\quad - \langle\langle Z_{j_1 \beta_1} Z_{j_1 \beta_2} Z_{j_2 \beta_3} Z_{j_2 \beta_4} \rangle\rangle_g \langle\langle Z_{i_1 \alpha_1} Z_{i_2 \alpha_2} \rangle\rangle_g] \\ &\quad + O(v^2) \end{aligned}$$

$$\begin{aligned} \bullet \langle\langle Z_{i_1 \alpha_1} Z_{i_2 \alpha_2} \rangle\rangle_g &= \int \frac{\prod_{i, \alpha} dz_{i\alpha}}{|2\pi g|^{1/2}} e^{(-\frac{1}{2} \sum_{j, \beta_1, \beta_2} Z_{j\beta_1} g^{\beta_1 \beta_2} Z_{j\beta_2})} Z_{i_1 \alpha_1} Z_{i_2 \alpha_2} \\ &= \mathbb{E}_g[Z_{i_1 \alpha_1} Z_{i_2 \alpha_2}] \\ &= \delta_{i_1 i_2} g_{\alpha_1 \alpha_2} \end{aligned}$$

$$\begin{aligned} \bullet \langle\langle Z_{j_1 \beta_1} Z_{j_1 \beta_2} Z_{j_2 \beta_3} Z_{j_2 \beta_4} \rangle\rangle_g &= \mathbb{E}[Z_{j_1 \beta_1} Z_{j_1 \beta_2} Z_{j_2 \beta_3} Z_{j_2 \beta_4}] \\ &= \mathbb{E}[Z_{j_1 \beta_1} Z_{j_1 \beta_2}] \mathbb{E}[Z_{j_2 \beta_3} Z_{j_2 \beta_4}] \\ &\quad + \mathbb{E}[Z_{j_1 \beta_1} Z_{j_2 \beta_3}] \mathbb{E}[Z_{j_1 \beta_2} Z_{j_2 \beta_4}] \\ &\quad + \mathbb{E}[Z_{j_1 \beta_1} Z_{j_2 \beta_4}] \mathbb{E}[Z_{j_1 \beta_2} Z_{j_2 \beta_3}] \\ &= g_{\beta_1 \beta_2} g_{\beta_3 \beta_4} + \delta_{j_1 j_2} g_{\beta_1 \beta_3} g_{\beta_2 \beta_4} + \delta_{j_1 j_2} g_{\beta_1 \beta_4} g_{\beta_2 \beta_3} \end{aligned}$$

$$\begin{aligned} \bullet \langle\langle Z_{i_1 \alpha_1} Z_{i_2 \alpha_2} \rangle\rangle_g \langle\langle Z_{j_1 \beta_1} Z_{j_1 \beta_2} Z_{j_2 \beta_3} Z_{j_2 \beta_4} \rangle\rangle_g &= \\ &= \delta_{i_1 i_2} g_{\alpha_1 \alpha_2} g_{\beta_1 \beta_2} g_{\beta_3 \beta_4} + \delta_{i_1 i_2} \delta_{j_1 j_2} g_{\beta_1 \beta_3} g_{\beta_2 \beta_4} g_{\alpha_1 \alpha_2} + \delta_{i_1 i_2} \delta_{j_1 j_2} g_{\alpha_1 \alpha_2} g_{\beta_1 \beta_4} g_{\beta_2 \beta_3} \end{aligned}$$

$$\langle Z_{j_1 \beta_1} Z_{j_2 \beta_2} Z_{j_3 \beta_3} Z_{j_4 \beta_4} Z_{i_1 \alpha_1} Z_{i_2 \alpha_2} \rangle := \langle Z_1 Z_2 Z_3 Z_4 Z_5 Z_6 \rangle_g$$

$$= E[Z_1 Z_2] E[Z_3 Z_4] E[Z_5 Z_6] + E[Z_1 Z_2] E[Z_3 Z_5] E[Z_4 Z_6] + E[Z_1 Z_2] E[Z_3 Z_6] E[Z_4 Z_5] \\ + E[Z_1 Z_3] E[Z_2 Z_4] E[Z_5 Z_6] + E[Z_1 Z_3] E[Z_2 Z_5] E[Z_4 Z_6] + E[Z_1 Z_3] E[Z_2 Z_6] E[Z_4 Z_5] \\ + E[Z_1 Z_4] [E[Z_2 Z_3] E[Z_5 Z_6] + E[Z_2 Z_5] E[Z_3 Z_6] + E[Z_2 Z_6] E[Z_3 Z_5]] \\ + E[Z_1 Z_5] [E[Z_2 Z_3] E[Z_4 Z_6] + E[Z_2 Z_4] E[Z_3 Z_6] + E[Z_2 Z_6] E[Z_3 Z_4]] \\ + E[Z_1 Z_6] [E[Z_2 Z_3] E[Z_4 Z_5] + E[Z_2 Z_4] E[Z_3 Z_5] + E[Z_2 Z_5] E[Z_3 Z_4]]$$

$$= \delta_{j_1 j_1} g_{\beta_1 \beta_2} [\delta_{j_2 j_2} g_{\beta_3 \beta_4} \delta_{i_1 i_2} g_{\alpha_1 \alpha_2} + \delta_{j_2 i_1} g_{\beta_3 \alpha_1} \delta_{j_2 i_2} g_{\beta_4 \alpha_2} + \delta_{j_2 i_2} g_{\beta_3 \alpha_1} \delta_{j_2 i_1} g_{\beta_4 \alpha_2}] \\ + \delta_{j_1 j_2} g_{\beta_1 \beta_3} [\delta_{j_3 j_3} g_{\beta_2 \beta_4} \delta_{i_1 i_2} g_{\alpha_1 \alpha_2} + \delta_{j_3 i_1} g_{\beta_2 \alpha_1} \delta_{j_3 i_2} g_{\beta_4 \alpha_2} + \delta_{j_3 i_2} g_{\beta_2 \alpha_2} + \delta_{j_3 i_1} g_{\beta_4 \alpha_1}] \\ + \delta_{j_1 j_3} g_{\beta_1 \beta_4} [\delta_{j_4 j_4} g_{\beta_2 \beta_3} \delta_{i_1 i_2} g_{\alpha_1 \alpha_2} + \delta_{j_4 i_1} g_{\beta_2 \alpha_1} \delta_{j_4 i_2} g_{\beta_3 \alpha_1} + \delta_{j_4 i_2} g_{\beta_2 \alpha_2} \delta_{j_4 i_1} g_{\beta_3 \alpha_1}] \\ + \delta_{j_1 i_1} g_{\beta_1 \alpha_1} [\delta_{j_2 j_2} g_{\beta_2 \beta_3} \delta_{j_3 i_2} g_{\beta_4 \alpha_2} + \delta_{j_2 j_3} g_{\beta_2 \beta_4} \delta_{j_3 i_2} g_{\beta_3 \alpha_2} + \delta_{j_2 i_2} g_{\beta_2 \alpha_2} \delta_{j_3 j_3} g_{\beta_3 \beta_4}] \\ + \delta_{j_1 i_2} g_{\beta_1 \alpha_2} [\delta_{j_2 j_2} g_{\beta_2 \beta_3} \delta_{j_3 i_1} g_{\beta_4 \alpha_1} + \delta_{j_2 j_3} g_{\beta_2 \beta_4} \delta_{j_3 i_1} g_{\beta_3 \alpha_1} + \delta_{j_2 i_1} g_{\beta_2 \alpha_1} \delta_{j_3 j_3} g_{\beta_3 \beta_4}]$$

//

$$\langle Z_{j_1 \beta_1} Z_{j_2 \beta_2} Z_{j_3 \beta_3} Z_{j_4 \beta_4} Z_{i_1 \alpha_1} Z_{i_2 \alpha_2} \rangle_g \\ - \langle Z_{i_1 \alpha_1} Z_{i_2 \alpha_2} \rangle_g \langle Z_{j_1 \beta_1} Z_{j_2 \beta_2} Z_{j_3 \beta_3} Z_{j_4 \beta_4} \rangle_g =$$

$$= \delta_{j_1 j_1} g_{\beta_1 \beta_2} [\delta_{j_2 j_2} g_{\beta_3 \beta_4} \delta_{i_1 i_2} g_{\alpha_1 \alpha_2} + \delta_{j_2 i_1} g_{\beta_3 \alpha_1} \delta_{j_2 i_2} g_{\beta_4 \alpha_2} + \delta_{j_2 i_2} g_{\beta_3 \alpha_1} \delta_{j_2 i_1} g_{\beta_4 \alpha_2}] \\ + \delta_{j_1 j_2} g_{\beta_1 \beta_3} [\delta_{j_3 j_3} g_{\beta_2 \beta_4} \delta_{i_1 i_2} g_{\alpha_1 \alpha_2} + \delta_{j_3 i_1} g_{\beta_2 \alpha_1} \delta_{j_3 i_2} g_{\beta_4 \alpha_2} + \delta_{j_3 i_2} g_{\beta_2 \alpha_2} + \delta_{j_3 i_1} g_{\beta_4 \alpha_1}] \\ + \delta_{j_1 j_3} g_{\beta_1 \beta_4} [\delta_{j_4 j_4} g_{\beta_2 \beta_3} \delta_{i_1 i_2} g_{\alpha_1 \alpha_2} + \delta_{j_4 i_1} g_{\beta_2 \alpha_1} \delta_{j_4 i_2} g_{\beta_3 \alpha_1} + \delta_{j_4 i_2} g_{\beta_2 \alpha_2} \delta_{j_4 i_1} g_{\beta_3 \alpha_1}] \\ + \delta_{j_1 i_1} g_{\beta_1 \alpha_1} [\delta_{j_2 j_2} g_{\beta_2 \beta_3} \delta_{j_3 i_2} g_{\beta_4 \alpha_2} + \delta_{j_2 j_3} g_{\beta_2 \beta_4} \delta_{j_3 i_2} g_{\beta_3 \alpha_2} + \delta_{j_2 i_2} g_{\beta_2 \alpha_2} \delta_{j_3 j_3} g_{\beta_3 \beta_4}] \\ + \delta_{j_1 i_2} g_{\beta_1 \alpha_2} [\delta_{j_2 j_2} g_{\beta_2 \beta_3} \delta_{j_3 i_1} g_{\beta_4 \alpha_1} + \delta_{j_2 j_3} g_{\beta_2 \beta_4} \delta_{j_3 i_1} g_{\beta_3 \alpha_1} + \delta_{j_2 i_1} g_{\beta_2 \alpha_1} \delta_{j_3 j_3} g_{\beta_3 \beta_4}]$$

$$- [\delta_{i_1 i_2} g_{\alpha_1 \alpha_2} g_{\beta_1 \beta_2} g_{\beta_3 \beta_4} + \delta_{i_1 i_2} \delta_{j_1 j_2} g_{\beta_1 \beta_3} g_{\beta_2 \beta_4} g_{\alpha_1 \alpha_2} + \delta_{i_1 i_2} \delta_{j_1 j_2} g_{\alpha_1 \alpha_2} g_{\beta_1 \beta_2}]$$

Na fórmula do livro

e as $\delta_{j_1 i_1}$?

$$E[Z_{i_1 \alpha_1} Z_{i_2 \alpha_2}] = \delta_{i_1 i_2} [g_{\alpha_1 \alpha_2} + \frac{1}{2} \sum_{\beta_1, \beta_2, \beta_3, \beta_4} \sqrt{(\beta_1 \beta_2)(\beta_3 \beta_4)} (g_{\alpha_1 \beta_1} g_{\alpha_2 \beta_2} g_{\beta_3 \beta_4} - g_{\alpha_1 \beta_1} g_{\alpha_2 \beta_3} g_{\beta_2 \beta_4})]$$

SEÇÃO Algebraic Deviation

Obs: Até dizer o contrário, daqui em diante tudo é na 2ª camada, i.e., $Z=Z^{(2)}$, $G=G^{(2)}$.

$$|2\pi \hat{G}|^{\frac{n_2}{2}} = \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T \hat{G}^{-1} Z\right)$$

$$= \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right)$$

$$\times \left\{ 1 + \frac{1}{2} Z^T G^{-1} \hat{\Delta} G G^{-1} Z - \frac{1}{2} Z^T G^{-1} \hat{\Delta} G G^{-1} \hat{\Delta} G G^{-1} Z + \frac{1}{2!} \left(\frac{1}{2}\right)^2 (Z^T G^{-1} \hat{\Delta} G G^{-1} Z)^2 + \mathcal{O}(\Delta^3) \right\}$$

$$= \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right)$$

$$+ \frac{1}{2} \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right) \frac{1}{2} Z^T G^{-1} \hat{\Delta} G G^{-1} Z$$

$$- \frac{1}{2} \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right) Z^T G^{-1} \hat{\Delta} G G^{-1} \hat{\Delta} G G^{-1} Z$$

$$+ \frac{1}{8} \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right) (Z^T G^{-1} \hat{\Delta} G G^{-1} Z)^2$$

$$= \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right)$$

$$+ \frac{1}{2} \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right) \left[\sum_{i=1}^{n_2} \sum_{\alpha_1, \alpha_2} \left(\sum_{\beta_1, \beta_2} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1 \beta_2} G^{\beta_2 \alpha_2} \right) Z_{i, \alpha_1} Z_{i, \alpha_2} \right]$$

$$- \frac{1}{2} \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right) \left[\sum_{i=1}^{n_2} \sum_{\alpha_1, \alpha_2} \left(\sum_{\beta_1, \beta_2} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1 \beta_2} G^{\beta_2 \alpha_2} \hat{\Delta} G_{\beta_2 \beta_3} G^{\beta_3 \alpha_3} \right) Z_{i, \alpha_1} Z_{i, \alpha_2} \right]$$

$$+ \frac{1}{8} \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right) \left[\sum_{i=1}^{n_2} \sum_{\alpha_1, \alpha_2, \alpha_3, \alpha_4} G^{\alpha_1 \beta_1} \dots G^{\alpha_4 \beta_4} \hat{\Delta} G_{\beta_1 \beta_2} \hat{\Delta} G_{\beta_3 \beta_4} Z_{i, \alpha_1} Z_{i, \alpha_2} Z_{i, \alpha_3} Z_{i, \alpha_4} \right]$$

$$+ \frac{1}{8} \int \prod_{i,\alpha} [\pi dz_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-L} Z\right) \mathcal{O}(\Delta^3)$$

$$= \sqrt{2\pi G}^{n_2} \left[1 + \frac{n_2}{2} \sum_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_1 \beta_2} G^{\beta_1 \beta_2} + \sum_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_1 \beta_2} \hat{\Delta} G_{\beta_2 \beta_1} \left(\frac{n_2^2}{8} G^{\beta_1 \beta_2} G^{\beta_2 \beta_1} - \frac{n_2}{4} G^{\beta_1 \beta_2} G^{\beta_2 \beta_1} \right) + \mathcal{O}(\Delta^3) \right]$$

$$\int_{i, \alpha} [\pi dz_{i, \alpha}] \exp(-\frac{1}{2} z^T G^{-1} z) = \sqrt{|2\pi G|^{n_2}}$$

$$\begin{aligned} & \frac{1}{2} \int_{i, \alpha} [\pi dz_{i, \alpha}] \exp(-\frac{1}{2} z^T G^{-1} z) \left[\sum_{i=1}^{n_1} \sum_{\alpha_1, \alpha_2} \left(\sum_{\beta_1, \beta_2} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2 \alpha_2} \right) z_{i, \alpha_1}^{(1)} z_{i, \alpha_2}^{(2)} \right] \\ &= \frac{1}{2} \sum_{i=1}^{n_1} \sum_{\alpha_1, \alpha_2} \int_{i, \alpha} [\pi dz_{i, \alpha}] \exp(-\frac{1}{2} z^T G^{-1} z) \left(\sum_{\beta_1, \beta_2} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2 \alpha_2} \right) z_{i, \alpha_1}^{(1)} z_{i, \alpha_2}^{(2)} \\ &= \frac{1}{2} \sum_{i=1}^{n_1} \sum_{\alpha_1, \alpha_2} \left(\sum_{\beta_1, \beta_2} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2 \alpha_2} \right) \int_{i, \alpha} [\pi dz_{i, \alpha}] \exp(-\frac{1}{2} z^T G^{-1} z) z_{i, \alpha_1}^{(1)} z_{i, \alpha_2}^{(2)} \\ &= \frac{1}{2} \sum_{i=1}^{n_1} \sum_{\alpha_1, \alpha_2} \left(\sum_{\beta_1, \beta_2} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2 \alpha_2} \right) |2\pi G|^{\frac{n_2}{2}} \mathbb{E}[z_{i, \alpha_1}^{(1)} z_{i, \alpha_2}^{(2)}] \\ &= \frac{1}{2} n_1 |2\pi G|^{\frac{n_2}{2}} \sum_{\alpha_1, \alpha_2} \left(\sum_{\beta_1, \beta_2} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2 \alpha_2} \right) G_{\alpha_2 \alpha_1} \\ &= \frac{1}{2} n_1 |2\pi G|^{\frac{n_2}{2}} \sum_{\alpha_1, \alpha_2=1}^4 \left(\sum_{\beta_1, \beta_2} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2 \alpha_2} \right) G_{\alpha_2 \alpha_1} \\ &= \frac{1}{2} n_1 |2\pi G|^{\frac{n_2}{2}} \sum_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_1 \beta_2} \end{aligned}$$

$$\begin{aligned} & \sum_{\alpha_1, \alpha_2=1}^4 \left(\sum_{\beta_1, \beta_2} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2 \alpha_2} \right) G_{\alpha_2 \alpha_1} \\ &= \sum_{\beta_1, \beta_2} \sum_{\alpha_1, \alpha_2} G_{\alpha_2 \alpha_1} G^{\alpha_1 \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2 \alpha_2} \\ &= \sum_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_1, \beta_2} \sum_{\alpha_1=1}^n G^{\alpha_1 \beta_1} \sum_{\alpha_2=1}^n G_{\alpha_1 \alpha_2} G^{\alpha_2 \beta_2} \\ &= \sum_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_1, \beta_2} \sum_{\alpha_1=1}^n G^{\alpha_1 \beta_1} \delta_{\alpha_1 \beta_2} \\ &= \sum_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_1 \beta_2} \end{aligned}$$

$$-\frac{1}{2} \int \prod_{i,\alpha} [dZ_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-1} Z\right) \left[\sum_{i=1}^{n_2} \sum_{\alpha_1, \alpha_2} \left(\sum_{\beta_1, \beta_2} G^{\alpha_1, \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2, \beta_3} \hat{\Delta} G_{\beta_3, \beta_4} G^{\beta_4, \alpha_2} \right) Z_{i, \alpha_1} Z_{i, \alpha_2} \right]$$

$$= -\frac{1}{2} \sum_{\alpha_1, \alpha_2} \left(\sum_{\beta_1, \beta_2} G^{\alpha_1, \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2, \beta_3} \hat{\Delta} G_{\beta_3, \beta_4} G^{\beta_4, \alpha_2} \right) \sum_{i=1}^{n_2} \int \prod_{i,\alpha} [dZ_{i,\alpha}] \exp\left(-\frac{1}{2} Z^T G^{-1} Z\right) Z_{i, \alpha_1} Z_{i, \alpha_2}$$

$$= -\frac{1}{2} \sum_{\alpha_1, \alpha_2} \left(\sum_{\beta_1, \beta_2} G^{\alpha_1, \beta_1} \hat{\Delta} G_{\beta_1, \beta_2} G^{\beta_2, \beta_3} \hat{\Delta} G_{\beta_3, \beta_4} G^{\beta_4, \alpha_2} \right) \eta_2 G_{\alpha_1, \alpha_2}$$

$$= -\frac{n_2}{2} \sum_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_3, \beta_4} G^{\beta_2, \beta_3} \sum_{\alpha_1} G^{\alpha_1, \beta_1} \sum_{\alpha_2} G^{\beta_4, \alpha_2} G_{\alpha_2, \alpha_1}$$

\$\delta_{\beta_4, \alpha_1}\$

$$= -\frac{n_2}{2} \sum_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_3, \beta_4} G^{\beta_2, \beta_3} \sum_{\alpha_1} G^{\alpha_1, \beta_1} \delta_{\beta_4, \alpha_1}$$

$$= -\frac{n_2}{2} \sum_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_1, \beta_2} \hat{\Delta} G_{\beta_3, \beta_4} G^{\beta_2, \beta_3} G^{\beta_1, \beta_4}$$

$$+ \frac{1}{8} \int [\prod dz_{i,\alpha}] \exp(-\frac{1}{2} z^T G^{-1} z) \left[\sum_{i_1, i_2}^{\alpha_2} \sum_{\alpha_1, \dots, \alpha_4} G^{\alpha_1 \beta_1} \dots G^{\alpha_4 \beta_4} \hat{G}_{\beta_1 \beta_2} \hat{G}_{\beta_3 \beta_4} z_{i_1 \alpha_1} z_{i_1 \alpha_2} z_{i_2 \alpha_3} z_{i_2 \alpha_4} \right]$$

$$= \sum_{\alpha_1, \dots, \beta_4} \hat{G}_{\beta_1 \beta_2} \hat{G}_{\beta_3 \beta_4} G^{\alpha_1 \beta_1} \dots G^{\alpha_4 \beta_4} \sum_{i_1, i_2} \int [\prod dz_{i,\alpha}] \exp(-\frac{1}{2} z^T G^{-1} z) z_{i_1 \alpha_1} z_{i_1 \alpha_2} z_{i_2 \alpha_3} z_{i_2 \alpha_4}$$

$$= \sum_{\alpha_1, \dots, \beta_4} \hat{G}_{\beta_1 \beta_2} \hat{G}_{\beta_3 \beta_4} G^{\alpha_1 \beta_1} \dots G^{\alpha_4 \beta_4} \left[\sum_{i_1 \neq i_2} \int [\prod dz_{i,\alpha}] \exp(-\frac{1}{2} z^T G^{-1} z) z_{i_1 \alpha_1} z_{i_1 \alpha_2} z_{i_2 \alpha_3} z_{i_2 \alpha_4} + \sum_i \int [\prod dz_{i,\alpha}] \exp(-\frac{1}{2} z^T G^{-1} z) z_{i \alpha_1} z_{i \alpha_2} z_{i \alpha_3} z_{i \alpha_4} \right]$$

$$= \sum_{\alpha_1, \dots, \beta_4} \hat{G}_{\beta_1 \beta_2} \hat{G}_{\beta_3 \beta_4} G^{\alpha_1 \beta_1} \dots G^{\alpha_4 \beta_4} \left[\sum_{i_1 \neq i_2} G_{\alpha_1 \alpha_2} G_{\alpha_3 \alpha_4} + n \mathbb{E}[z_{i \alpha_1} z_{i \alpha_2} z_{i \alpha_3} z_{i \alpha_4}] \right]$$

$$= \sum_{\alpha_1, \dots, \beta_4} \hat{G}_{\beta_1 \beta_2} \hat{G}_{\beta_3 \beta_4} G^{\alpha_1 \beta_1} \dots G^{\alpha_4 \beta_4} \left[(n^2 - n) G_{\alpha_1 \alpha_2} G_{\alpha_3 \alpha_4} + n (G_{\alpha_1 \alpha_2} G_{\alpha_3 \alpha_4} + G_{\alpha_1 \alpha_3} G_{\alpha_2 \alpha_4} + G_{\alpha_1 \alpha_4} G_{\alpha_2 \alpha_3}) \right]$$

$$= \sum_{\alpha_1, \dots, \beta_4} \hat{G}_{\beta_1 \beta_2} \hat{G}_{\beta_3 \beta_4} G^{\alpha_1 \beta_1} \dots G^{\alpha_4 \beta_4} \left[n^2 G_{\alpha_1 \alpha_2} G_{\alpha_3 \alpha_4} + n (G_{\alpha_1 \alpha_3} G_{\alpha_2 \alpha_4} + G_{\alpha_1 \alpha_4} G_{\alpha_2 \alpha_3}) \right]$$

$$= \sum_{\beta_1, \dots, \beta_4} \hat{G}_{\beta_1 \beta_2} \hat{G}_{\beta_3 \beta_4} \sum_{\alpha_1} G^{\alpha_1 \beta_1} \sum_{\alpha_2} G^{\alpha_2 \beta_2} \sum_{\alpha_3} G^{\alpha_3 \beta_3} \sum_{\alpha_4} G^{\alpha_4 \beta_4} \left[n^2 G_{\alpha_1 \alpha_2} G_{\alpha_3 \alpha_4} + n (G_{\alpha_1 \alpha_3} G_{\alpha_2 \alpha_4} + G_{\alpha_1 \alpha_4} G_{\alpha_2 \alpha_3}) \right]$$

$$= \dots$$

Colocando de volta ^(4.56) e (4.57) na distribuição (4.53)
 Obs: até aqui, tudo tinha índice (2). Agora muda tudo.

$$P(z^{(2)} | D) = \int \frac{[\prod_{i,\alpha} dz_{i,\alpha}^{(1)}]}{(2\pi \hat{G}_1)^{\frac{n_2}{2}}} \exp\left[-\frac{1}{2} z^T \hat{G}_{(2)}^{-1} z\right] P(z^{(1)} | D)$$

$$= \int \frac{[\prod_{i,\alpha} dz_{i,\alpha}^{(1)}]}{(2\pi \hat{G}_1^{(2)})^{\frac{n_2}{2}}} \cdot \exp\left[-\frac{1}{2} z^T \hat{G}_{(2)}^{-1} z\right] \cdot \left\{ 1 + \frac{1}{2} z^T \hat{G}_1^{-1} \Delta \hat{G}_1 \hat{G}_1^{-1} z - \frac{1}{2} z^T \hat{G}_1^{-1} \Delta \hat{G}_1 \hat{G}_1^{-1} \Delta \hat{G}_1 \hat{G}_1^{-1} z + \frac{1}{2!} \left(\frac{1}{2}\right)^2 (z^T \hat{G}_1^{-1} \Delta \hat{G}_1 \hat{G}_1^{-1} z)^2 + O(\Delta^3) \right\}$$

$$\left[1 + \frac{n_2}{2} \sum_{p,p_2} \hat{\Delta G}_{1,p,p_2}^{(2)} G_{1,p,p_2}^{p,p_2} + \sum_{p,p_2} \hat{\Delta G}_{1,p,p_2}^{(2)} \hat{\Delta G}_{1,p,p_2}^{(2)} \left(\frac{n_2^2}{8} G_{1,p,p_2}^{p,p_2} G_{1,p,p_2}^{p,p_2} - \frac{n_2}{4} G_{1,p,p_2}^{p,p_2} G_{1,p,p_2}^{p,p_2} \right) + O(\Delta^3) \right]$$

cte
Depende de $z^{(1)}$
const.
lixo

$$\stackrel{(!)}{=} \frac{1}{(2\pi \hat{G}_1^{(2)})^{\frac{n_2}{2}}} \exp\left(-\frac{1}{2} \sum_{j=1}^{n_2} \sum_{\alpha_1, \alpha_2} G_{(2)}^{\alpha_1, \alpha_2} z_{j\alpha_1} z_{j\alpha_2}\right)$$

$$\propto \left\{ \left[1 + O\left(\frac{1}{n_1}\right) \right] + \sum_{i=1}^{n_2} \sum_{\alpha_1, \alpha_2} \left[O\left(\frac{1}{n_1}\right) \right] z_{i\alpha_1}^{(2)} z_{i\alpha_2}^{(2)} \right.$$

$$\left. + \frac{1}{8n_1} \sum_{i,j=1}^{n_2} \sum_{\alpha_1, \alpha_2, \alpha_3, \alpha_4} V_{(2)}^{(\alpha_1, \alpha_2)(\alpha_3, \alpha_4)} z_{i\alpha_1}^{(2)} z_{i\alpha_2}^{(2)} z_{j\alpha_3}^{(2)} z_{j\alpha_4}^{(2)} \right\}$$

Obs: Aqui diz pra usar $E[\hat{\Delta G}_{1,p_1,p_2}^{(2)}] = 0$ e $E[\hat{\Delta G}_{1,p_1,p_2}^{(2)} \hat{\Delta G}_{1,p_3,p_4}^{(2)}] = V_{(2)}^{(p_1,p_2)(p_3,p_4)}$.

Tomando o logaritmo (???) e observando o resto na part.

$$S(z) = \frac{1}{2} \sum_{\alpha_1, \alpha_2} \left[G_{(2)}^{\alpha_1, \alpha_2} + O\left(\frac{1}{n_1}\right) \right] \sum_i z_{i\alpha_1} z_{i\alpha_2}$$

$$- \frac{1}{8} \sum_{\alpha_1, \dots, \alpha_4} \frac{1}{n_1} V_{(2)}^{(\alpha_1, \alpha_2)(\alpha_3, \alpha_4)} \sum_{i,j} z_{i\alpha_1} z_{i\alpha_2} z_{j\alpha_3} z_{j\alpha_4} + O\left(\frac{1}{n_1^2}\right)$$